FUNCTIONAL SPECIFICATION · v1.0 · 20 OCTOBER 2026

Zarmeen Gardens Real Estate & Society ERP

How the system sells plots, collects installments, runs post-sale operations, bills and maintains the society, controls development cost and reports — on Frappe / ERPNext v16.

AT A GLANCE

One scheme, one ledger, every operation end to end

400
Kanal boundary

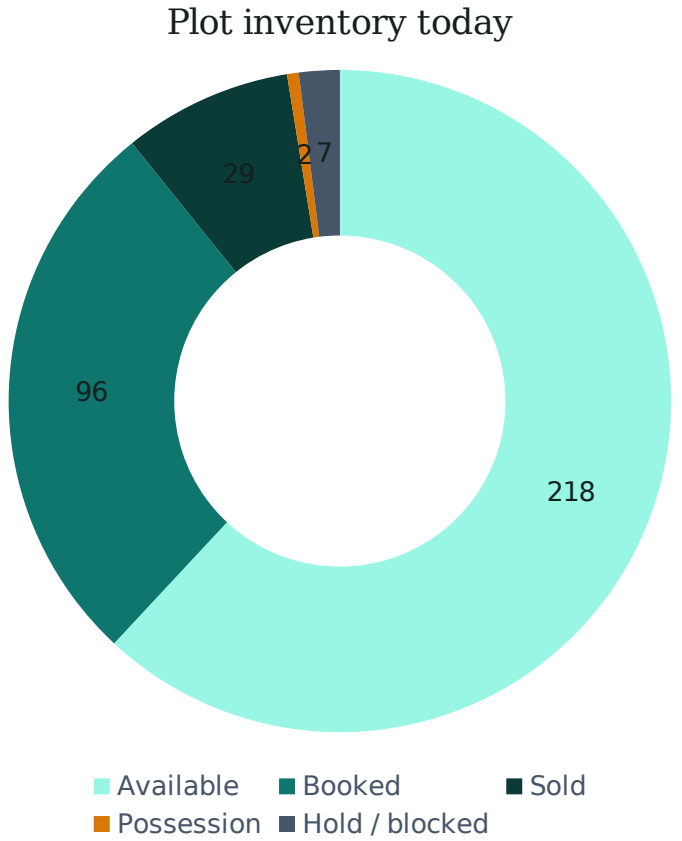
4
Blocks

352
Plots on the map

130
Bookings

16
Document types, EN / اردو

21
Reports with charts



Real-estate context around ERPNext accounting

In scope

- Land and map: boundary, blocks, plots, infrastructure, KML / GeoJSON import
- CRM and WhatsApp; booking and payment plans
- Installments, receipts and recovery
- Transfers, cancellations, re-allotments, merges, surcharge waivers
- Agent and dealer commissions
- Society billing, maintenance and complaints
- Development cost, procurement, stock, assets with depreciation
- Documents in English and Urdu; reports, dashboards, owner portal

Out of scope (this release)

- Online card payments for owners
- Construction (building) management after possession
- Live WhatsApp Business account — runs in mock mode until credentials are supplied
- Google Maps — supported, needs an API key

USERS AND ROLES

Eight roles, each scoped on the server

1 Admin

Setup, settings, users, imports, acceptance checks

Sees: Everything

2 Sales Manager

Bookings, discount and post-sale approvals, portal invites

Sees: All sales, CRM, collections

3 Sales Agent

Leads, deals, visits, bookings, receipts, chats; raises requests

Sees: Only their own records

4 Accounts

Receipts, invoices, refunds, commission and supplier payments

Sees: All financial documents

5 Recovery Officer

Recovery calls, promises, dunning, waiver requests

Sees: Overdue bookings, defaulters

6 Society Manager

Bills, bulk collection, charge heads, plans, merges

Sees: Society, maintenance, map

7 Projects

Development projects, budgets, material requests, tasks

Sees: Development, maintenance

8 Plot owner

Plots, schedule, bills, receipts, documents, complaints

Sees: Only their own, via the portal

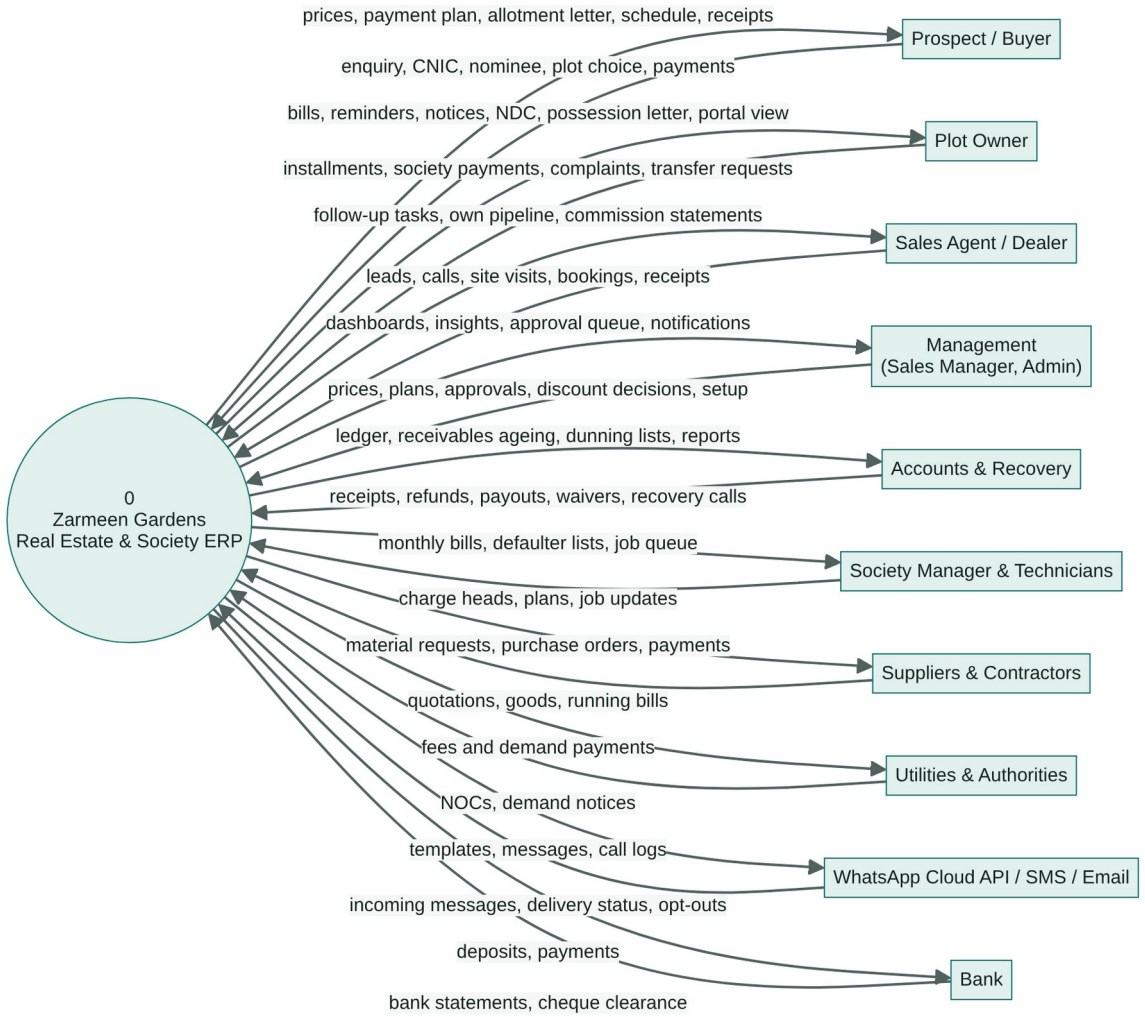
Data flow diagrams

Level 0 shows the system against the outside world. Level 1 breaks it into ten processes and ten shared data stores (D1–D10).

DFD

DFD LEVEL 0 · CONTEXT

The system as one process and the parties around it



How to read it

- Circle — the whole system, process 0
- Rectangle — an external entity: buyer, owner, agent, bank, supplier, contractor, regulator, WhatsApp
- Arrow — a data flow, labelled with what moves
- All stores live in the ERPNext database

Ten processes share ten data stores

1.0
Project & land setup
D1 D10

2.0
CRM & site visits
D2

3.0
Booking & schedule
D1 D3 D5

4.0
Collections & recovery
D4 D5

5.0
Post-sale operations
D1 D3 D5

6.0
Commissions
D5

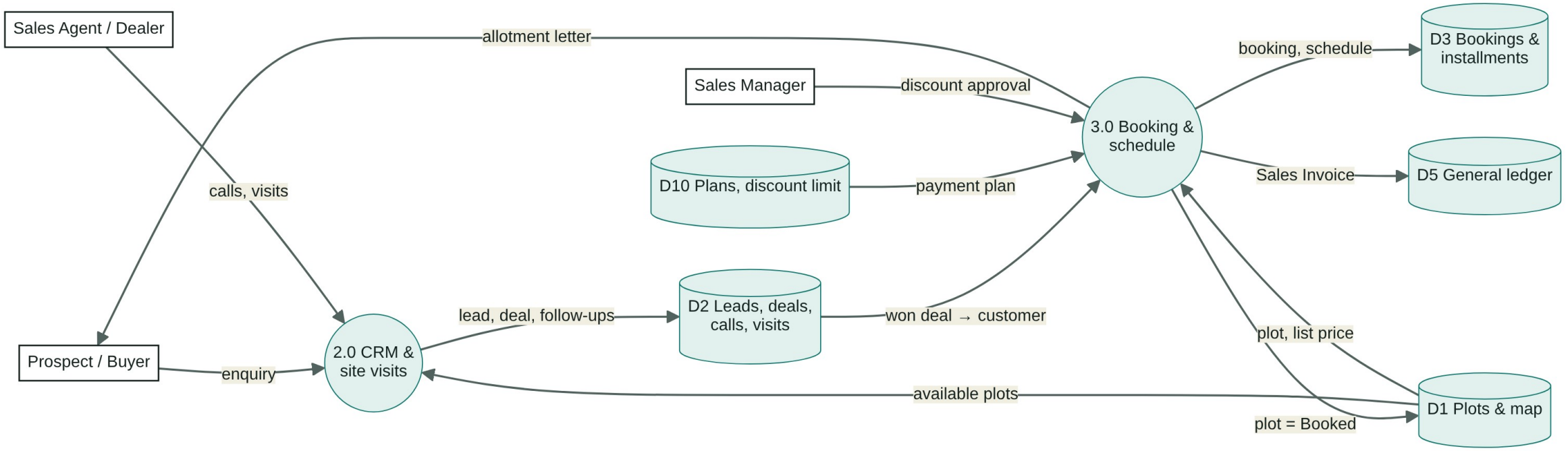
7.0
Society billing & maintenance
D5 D6 D7

8.0
Development, stock & assets
D5 D8

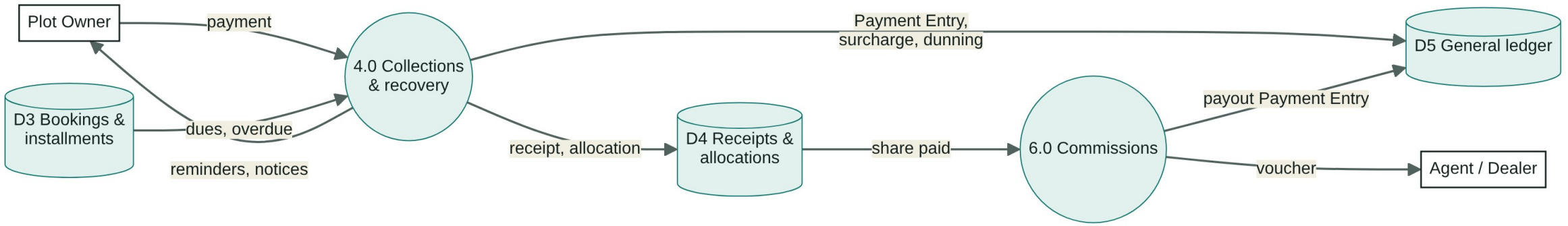
9.0
Documents & messaging
D9

10.0
Reports & dashboards
all

From enquiry to booked plot (2.0, 3.0)



Collections, recovery and commissions (4.0, 6.0)



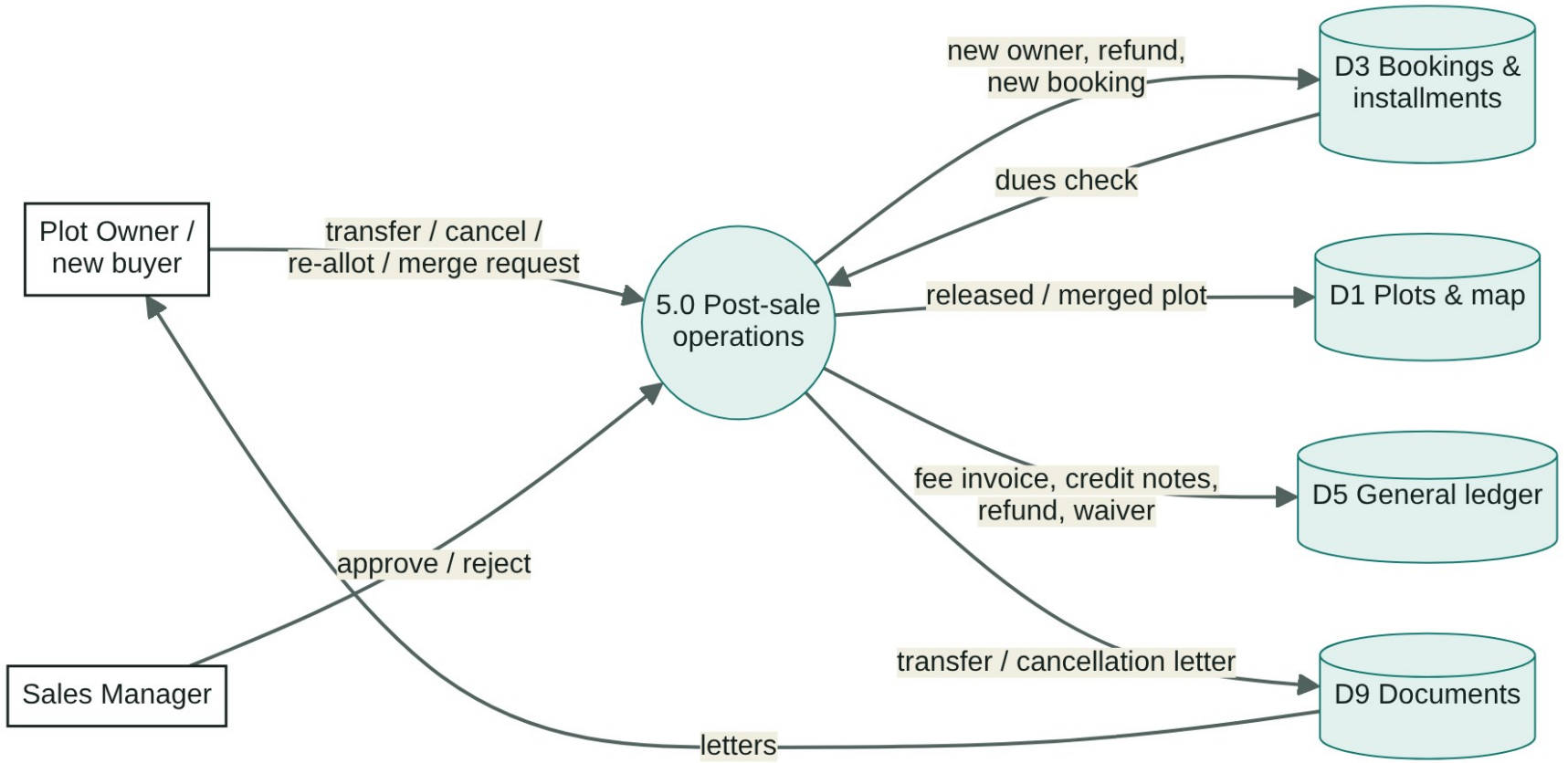
4.0 Receipts allocate oldest due first; surcharges and dunning post to the ledger

4.0 Reminders 3 days before due; overdue notices every 7 days

6.0 Commission becomes payable at 25% paid; payout by Payment Entry and voucher

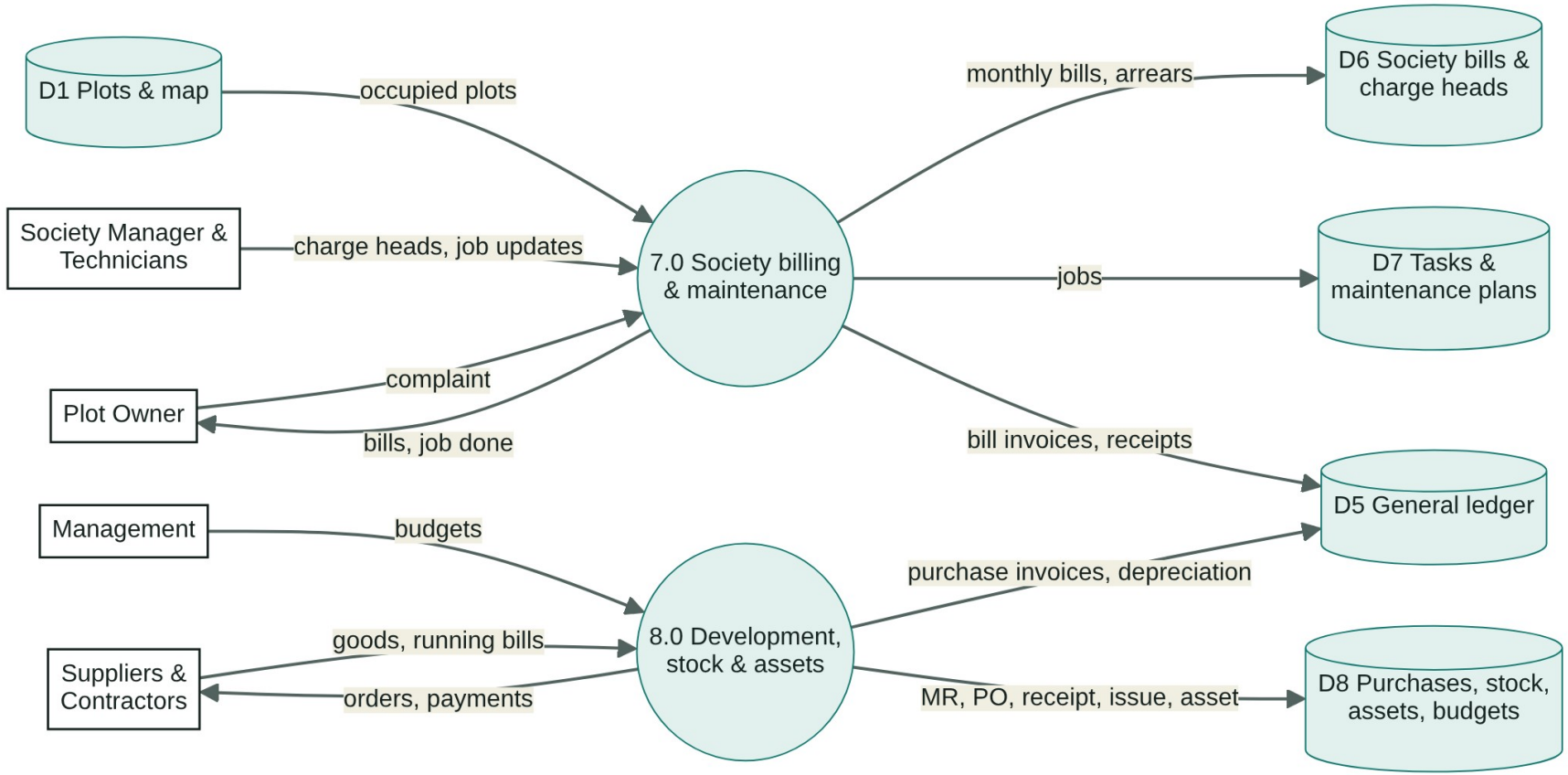
DFD LEVEL 1 · POST-SALE

Transfers, cancellations, re-allotments, merges (5.0)



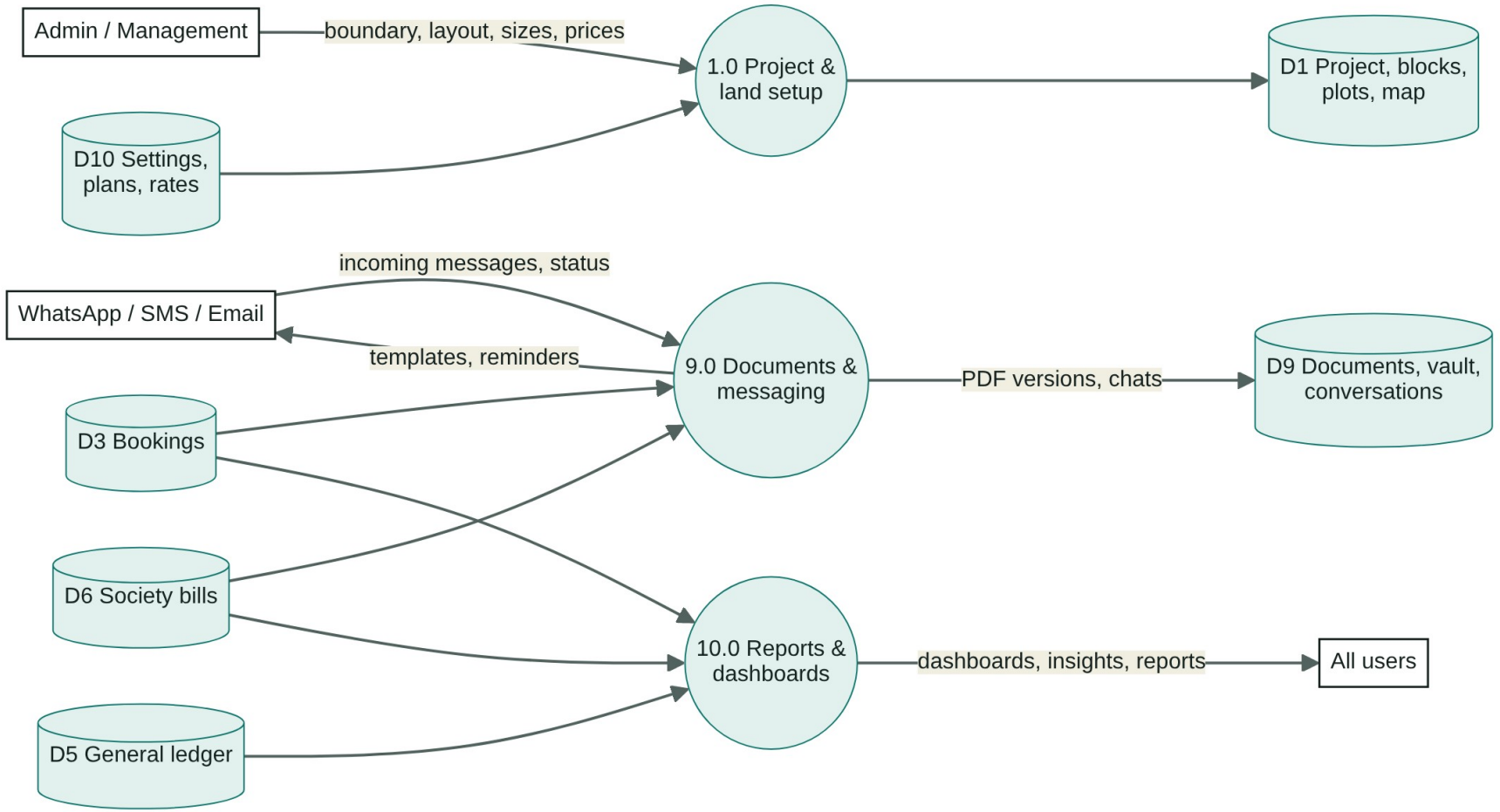
- Owner or buyer raises the request; the wizard checks dues first
- Sales Manager approves or rejects; Revise sends it back
- Approval posts fee invoices, credit notes, refunds or waivers
- Plots are released, re-allotted or merged on the map
- Transfer and cancellation letters are versioned in D9

Society, maintenance and development (7.0, 8.0)



- 7.0 bills every occupied plot on day 1, due in 15 days, arrears carried
- Complaints from portal, WhatsApp or staff become tasks; the owner is told when done
- 8.0 runs each cost head as a Project with a Budget
- Material Request → PO → Receipt → Invoice; issues charge the head
- Assets depreciate straight line by category

Setup, documents & messaging, reports (1.0, 9.0, 10.0)



- 1.0 builds the scheme: boundary, layout, sizes, prices
- 9.0 makes 16 document types as versioned PDFs, English or Urdu
- WhatsApp first, SMS / email fallback
- 10.0 reads every store for dashboards, insights and 21 reports

DATA STORES

D1–D10 are ordinary ERPNext and RE doctypes

Store	Holds	DocTypes
D1	Land & map	RE Project, RE Block, RE Plot (price history), RE Plot Size, RE Map Feature, RE Map Import, RE Land Use
D2	CRM	CRM Lead, CRM Deal, CRM Task, CRM Call Log, RE Site Visit
D3	Bookings	RE Booking (RE Installment rows), RE Payment Plan
D4	Receipts	RE Payment (RE Payment Allocation rows)
D5	Ledger	Sales Invoice, Payment Entry, Journal Entry, Dunning, GL Entry, RE Surcharge, RE Commission
D6	Society bills	RE Society Bill (RE Bill Charge rows), RE Charge Head (RE Charge Rate rows)
D7	Maintenance	Task, Project (maintenance), RE Maintenance Plan, ToDo
D8	Development	Material Request, Purchase Order / Receipt / Invoice, Stock Entry, Asset, Depreciation Schedule, Budget, Project
D9	Documents & chat	RE Generated Document, RE Vault Document, RE Conversation, WhatsApp Message, RE Reminder Log, RE Broadcast
D10	Settings	RE Settings, RE Payment Plan, RE Quick Reply, approval workflows

Functional requirements

What each process must do, the screen in the staff app and what it posts in ERPNext. The desk offers the same documents.

FR

FR-1 TO FR-3

Land, CRM and booking

1.0 Project & land

- Boundary drawn or imported (KML / GeoJSON); land use must total 100%
- Blocks, plot sizes with base price and maintenance; plots with flags and outline
- Price revisions kept as plot price history
- Infrastructure on the map; live map by status or payment health

2.0 CRM & site visits

- Leads from any source; an unknown WhatsApp number creates a lead
- Pipeline Opportunity → Site Visit → Negotiation → Booking creates the Customer
- Calls with outcome and promise to pay; missed calls create follow-ups
- Site visits with WhatsApp invitation and result

3.0 Booking & schedule

- Customer with CNIC and nominee, an available plot, price, discount, plan
- Discount above 5% waits for Sales Manager approval
- Submit builds the schedule, books the plot, posts a Sales Invoice with payment schedule
- Allotment letter and forms in English or Urdu

Collections, recovery and post-sale operations

4.0 Collections & recovery

- Receipts by cash, transfer or cheque, allocated oldest due first → Payment Entry
- Unpaid → Due (7 days) → Overdue, driven by the calendar
- Monthly late-payment surcharge after the grace period
- Recovery calls, promises, final notice, dunning
- Reminders 3 days before due; overdue notices every 7 days

Operation	Rule	Posting on approval
Transfer	No overdue installments or society dues; fee Rs 50,000	Fee invoice; credit note for seller; new invoice for buyer; transfer letter
Cancellation	Deduct 10% of price, never more than paid	Credit notes, refund Payment Entry, plot released, unpaid commission cancelled
Re-allotment	A plot freed by a cancellation; fee Rs 25,000	New booking and invoice; fee invoice
Merge	Adjacent available plots	New plot with combined outline, area and price
Surcharge waiver	Part or all, with a reason	Credit note against the surcharge invoice

FR-6 TO FR-8

Commissions, society and development

6.0 Commissions

- Agent 1%, dealer 2% of net price accrue at booking
- Payable once the customer has paid 25%
- Paid in batches by Payment Entry with a printable voucher; cancelling reverses it

7.0 Society

- Charge heads: fixed, per marla or by plot size
- Monthly bill per occupied plot, day 1, due in 15 days, with arrears
- Bulk collection by owner or block
- Complaints become tasks; plans create tasks ahead of due

8.0 Development

- Each cost head is a Project with a Budget
- Material Request → PO → Receipt → Invoice; Stock Entry issues to the head
- Contractor bills and NOC fees as Purchase Invoices
- Assets depreciated straight line by category

FR-9 AND FR-10

Documents, messaging, reports and portal

9.0 Documents & messaging

- 16 document types on the letterhead with a QR verification code
- Numbered PDF versions in English or Urdu (right to left)
- Vault for approvals, NOCs, registry, insurance with expiry alerts
- Shared WhatsApp inbox, templates, quick replies, opt-out keywords, rate-limited broadcasts

10.0 Reports, dashboards & portal

- Dashboard, Today, insights and a customer 360 with payer segment
- Report centre: preview, bar / line / area / pie / donut / trend, Excel and PDF
- Owner portal in English / اردو: plots on the map, schedule, bills, receipts, documents, complaints
- Staff preview the portal as an owner, read-only

Business rules and controls

§7

All money maths lives in one tested module (realestate/calc.py). Rates and fees are settings, not code — changes apply to new documents only.

BUSINESS RULES

Six calculations every document follows

Installment schedule

```
net = price − discount
fixed parts = net × % (round Rs 1,000)
each = (net − fixed) ÷ n; last absorbs the rest
Σ schedule = net, always exact
```

Receipt allocation

```
pays unpaid rows oldest due first
never more than total outstanding
```

Late-payment surcharge

```
after 15 days grace:
max(Rs 500, overdue × 2% / month)
capped at overdue × 10%
one per booking per month
```

Cancellation

```
deduction = min(net × 10%, paid)
refund = paid − deduction
```

Commission

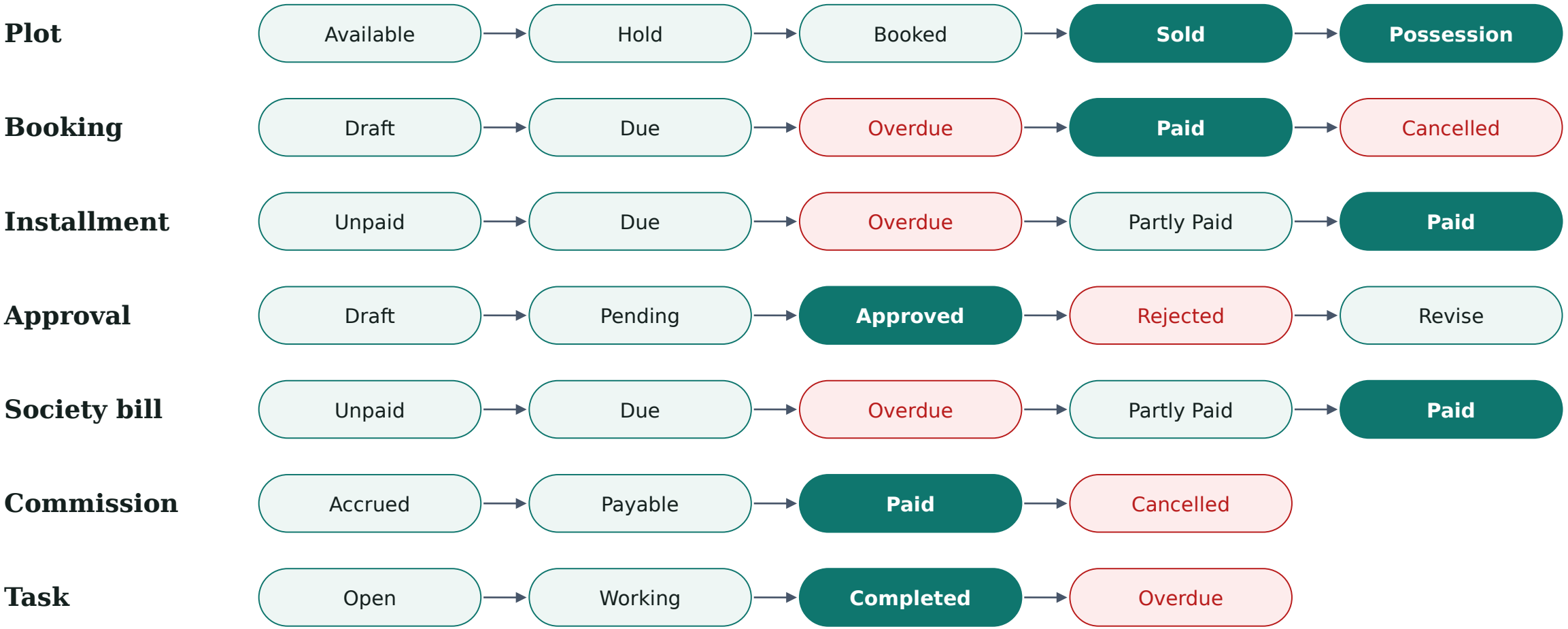
```
net × rate (agent 1%, dealer 2%)
Accrued → Payable at 25% paid → Paid
```

Society bill

```
Σ charge heads for the plot
(fixed | rate × marla | size rate)
payable = bill + arrears
```

STATUS MODELS

Every record moves through a fixed set of states



What runs on its own, and what the system talks to

Scheduled jobs

When	Job
Daily	Refresh dues · society bills on bill day · reminders · overdue notices · vault expiry
Daily	Surcharges · dunnings · commission statuses
Daily	Planned maintenance tasks · mark overdue jobs
Every minute	WhatsApp delivery status · SMS fallback
Month end	Asset depreciation
Demo only	A live day of demo activity (re-daily-data)

Integrations

Integration	Use	State
WhatsApp Cloud API	Templates, chat, delivery, webhook	Mock until credentials
SMS / email	Fallback when WhatsApp fails	Configured
Maps	MapLibre satellite; Google optional	Live; Google needs key
ERPNext	Accounts, buying, stock, assets, projects, dunning	Live

SECURITY, AUDIT AND QUALITY

Controls built into the platform

- ✓ **Role scoping**
  Agents see only their own records — enforced by server-side query conditions
- ✓ **Approvals**
  Transfers, cancellations, re-allotments, merges, waivers and large discounts
- ✓ **No deletion**
  Submitted financial documents are cancelled and amended, never deleted
- ✓ **Audit trail**
  Version history and an activity chain on every financial and settings record
- ✓ **Secrets**
  API tokens and map keys stored encrypted in RE Settings
- ✓ **Portal**
  Owners reach only their own records; staff preview is read-only

Non-functional requirements

Usability Phone width (390 px), light and dark; portal and documents in English or Urdu

Performance Lists and insights in about a second; previews up to 1,000 rows

Reliability Idempotent daily jobs; missed demo days caught up (up to 14)

Accuracy Schedules total the net price; ledger balances; 127 cross-page checks

Maintainability Standard ERPNext accounting; money maths in one tested module

Project and site setup

Nine steps from an empty bench to a live scheme, checked by the acceptance command.

§14

SETUP

Nine steps to go live

1 Install

ERPNext v16, Frappe CRM, WhatsApp; get-app and install-app realestate

2 Company & accounts

Fiscal year July–June, cost centers per block, banks, modes of payment

3 RE Settings

Discount limit, fees, surcharge and commission rules, billing day, WhatsApp, map

4 Project & map

Boundary, land use, blocks, plot sizes and prices, plots, infrastructure

5 Sales setup

Payment plans, agents and dealers, quick replies and templates

6 Society setup

Charge heads and size rates, maintenance plans and assignee

7 Development

Cost-head projects and budgets, site store, suppliers, asset categories

8 Users

Staff with roles; owner-portal invites from the customer page

9 Go live

Turn on the scheduler; re-acceptance until every check passes

ACCEPTANCE CRITERIA

Ten automated checks pass on the demo scheme

01	Land use totals 100% of the boundary (400.00 kanal)	PASS
02	Every plot inside its block, blocks inside the boundary, no overlaps	PASS
03	Every schedule = price − discount; invoice schedule = invoice total	PASS
04	Every receipt has a matching submitted Payment Entry; ledger balances	PASS
05	Map recolours live when a plot changes	PASS
06	Scheduled jobs enabled and current	PASS
07	Incoming WhatsApp reaches the inbox and timeline (mock)	PASS
08	All documents render to PDF on the letterhead, English and Urdu	PASS
09	Role scoping: agents, managers, portal owners	PASS
10	Audit trail and no deletion of financial records	PASS
11	Phone width, light and dark	BY EYE

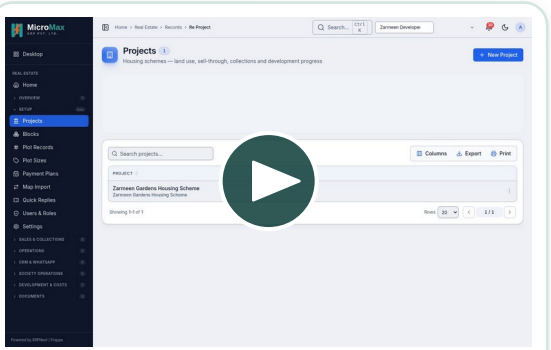
Simulator videos

Sixteen narrated screen recordings on the demo scheme — one per operation. Click a thumbnail to play it (files in the videos folder beside this deck).



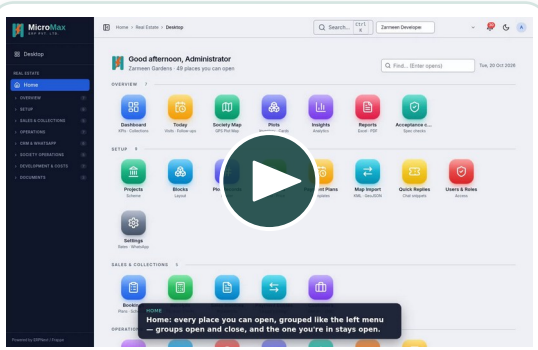
VIDEO LIBRARY

Setup, daily work and selling



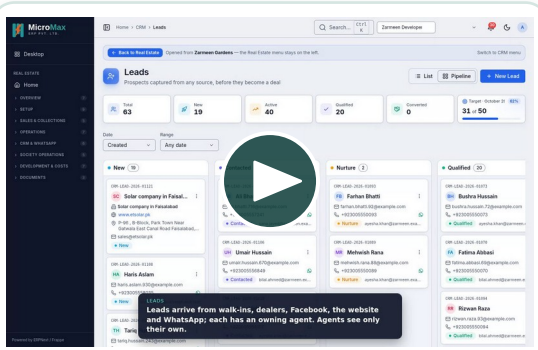
00 Project & site setup

Install, settings, project and map, plots, plans, users, acceptance



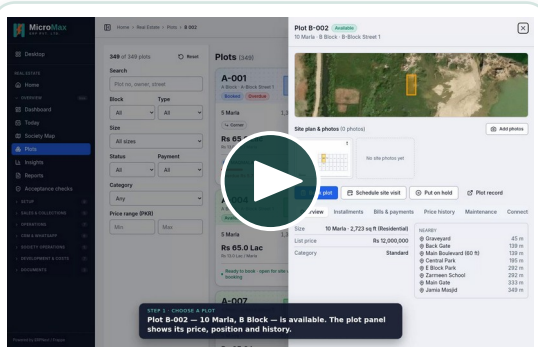
01 Your day in the app

Home, dashboard, today's visits and follow-ups, notifications



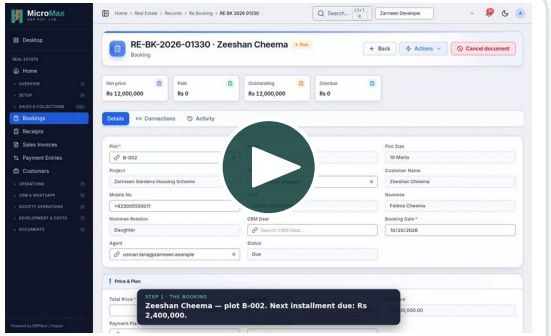
02 CRM & WhatsApp

Leads, deals, site visits, the shared inbox



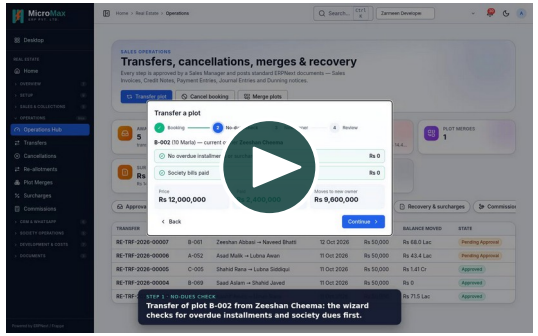
03 Booking a plot

Book, submit, schedule, invoice, Urdu allotment letter



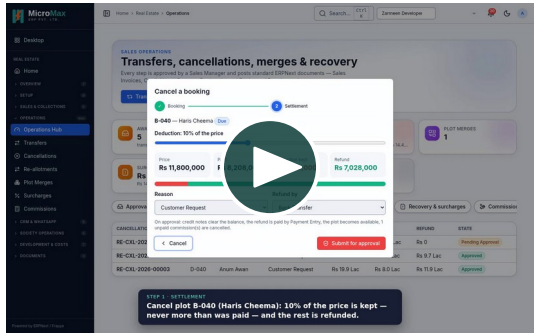
04 Receiving a payment

Oldest-due-first allocation, Payment Entry, ledger



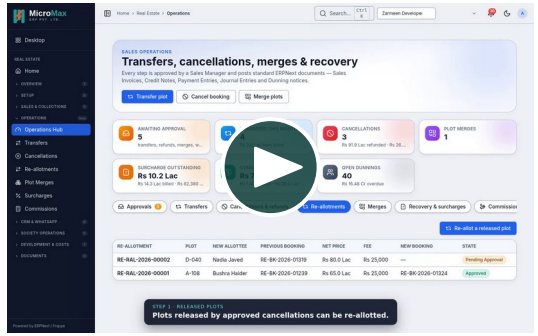
05 Transferring a plot

No-dues check, new owner, fee, approval



06 Cancelling a booking

Deduction and refund, approval, plot released

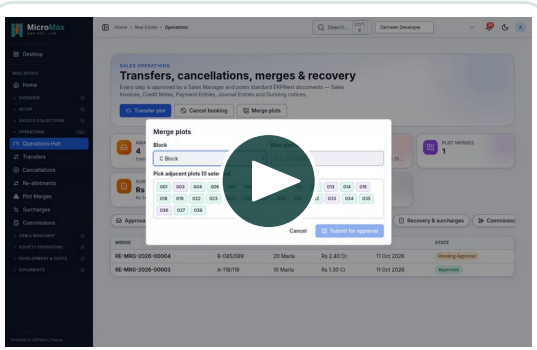


07 Re-allotting a plot

A freed plot goes to a waiting buyer

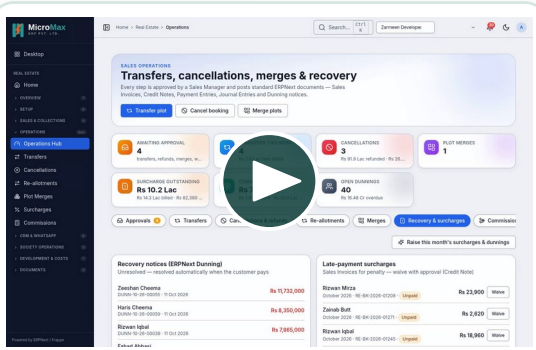
VIDEO LIBRARY

Post-sale, society, development and portal



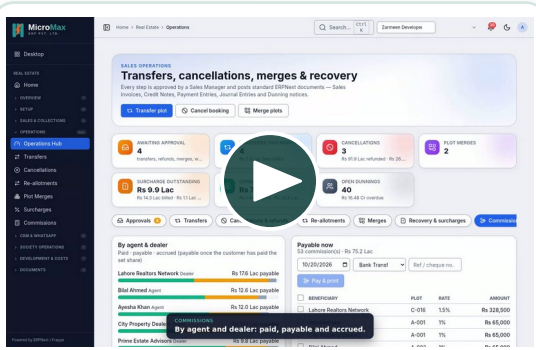
08 Merging plots

Two adjacent plots become one; the map updates



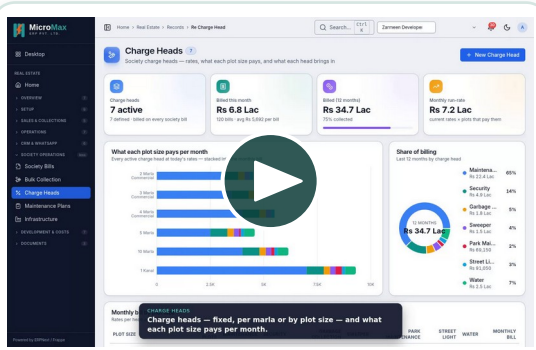
09 Recovery & surcharges

Dunning, monthly surcharges, a waiver



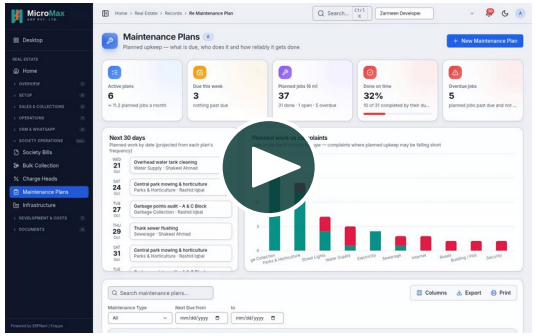
10 Commissions

Accrued, payable, paid with a voucher



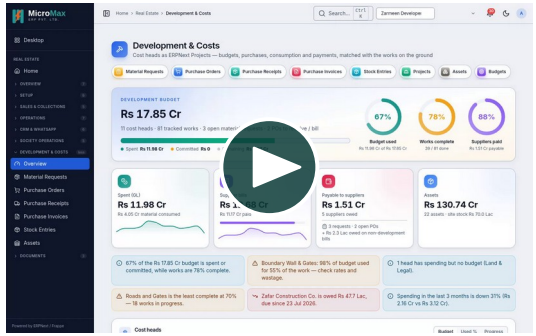
11 Society billing

Charge heads, bills with arrears, bulk collection



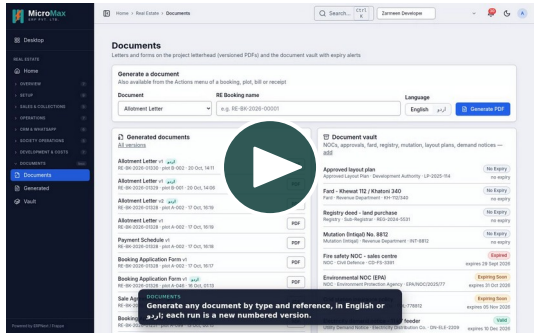
12 Maintenance

Plans, urgent queue, complaint jobs



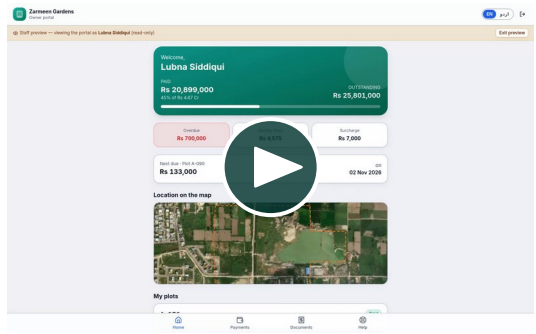
13 Development & assets

Budget vs actual, procurement, depreciation



14 Documents & reports

English / Urdu documents, report centre



15 Owner portal

What a plot owner sees, English and Urdu

ALL PHASES BUILT · ACCEPTANCE 10 / 10

Ready for real data

Open items need the owner: live WhatsApp Business credentials, a Google Maps key if wanted, and online card payments in a later release.